CLOSEDNESS OF THE LOCUS WHERE THE PICARD FIBRE IS TRIVIAL, FOR INTEGRAL GEOMETRIC FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9409-9416 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 9418-9427, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The Picard setup and characterization of Z (lemma-diagonal-picard-flat-proper, same chapter), the scheme-theoretic image reach-points criterion (Morphisms), and the triviality of invertible sheaves over valuation rings at the generic point (same chapter); cited inputs with the valuative argument written in the retained span.

Source reference. Original lemma 0BDP: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9199–9215. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0EX8: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9334–9343. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02JQ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9427–9427. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ and $\mathcal{E}$ be as in Lemma 0BDP and in addition assume $\mathcal{E}$ is an invertible $\mathcal{O}_X$-module. If moreover the geometric fibres of f are integral, then Z is closed in S.*

Proof. Since $j : Z \to S$ is of finite presentation, it suffices to show: for any morphism $g : \mathrm{Spec}(A) \to S$ where A is a valuation ring with fraction field K such that $g(\mathrm{Spec}(K)) \in j(Z)$ we have $g(\mathrm{Spec}(A)) \subset j(Z)$. See Morphisms, Lemma 02JQ. This follows from Lemma 0EX8 and the characterization of $j : Z \to S$ in Lemma 0BDP. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.